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*Course: Computer Systems Engineering*

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**Title: Tier-1 SOC Lab Report**

## Contents

|                                                      |           |
|------------------------------------------------------|-----------|
| <b>1. Introduction .....</b>                         | <b>3</b>  |
| <b>2. Objective .....</b>                            | <b>3</b>  |
| <b>3. Overview .....</b>                             | <b>3</b>  |
| <b>4. Step-by-Step Explanation.....</b>              | <b>3</b>  |
| <b>1. Lab Setup .....</b>                            | <b>3</b>  |
| <b>2. Event Generation .....</b>                     | <b>5</b>  |
| <b>3. Querying Logs .....</b>                        | <b>6</b>  |
| <b>4. Monitoring &amp; Triage .....</b>              | <b>7</b>  |
| <b>5. Enrichment.....</b>                            | <b>7</b>  |
| <b>5. Findings / Analysis .....</b>                  | <b>7</b>  |
| • <b>What happened?.....</b>                         | <b>7</b>  |
| • <b>Evidence .....</b>                              | <b>7</b>  |
| • <b>Actions Taken .....</b>                         | <b>8</b>  |
| • <b>Root Cause.....</b>                             | <b>8</b>  |
| • <b>Lessons Learned &amp; Prevention .....</b>      | <b>8</b>  |
| <b>6. Use-Case Sheet (Tier-1 Version) .....</b>      | <b>9</b>  |
| <b>7. Playbook Design Doc (Tier-1 Version) .....</b> | <b>9</b>  |
| <b>8. Shift Handover (Tier-1 Version).....</b>       | <b>10</b> |
| <b>9. Conclusion.....</b>                            | <b>10</b> |

## **1. Introduction**

- This lab simulates a Tier-1 SOC workflow using Azure Sentinel and SecurityEvent logs.
- Focus: monitoring, alert validation, first-look triage, and documenting incidents.
- Tools used: Log Analytics workspace, Azure Sentinel, Azure Monitor Agent (AMA), Windows VM, PowerShell.

## **2. Objective**

- To practice Tier-1 SOC monitoring and triage in a controlled lab environment.
- Generate events to simulate failed and successful logons.
- Learn to query, analyze, and document incidents using SecurityEvent logs.

## **3. Overview**

- A single VM is connected to a Log Analytics workspace.
- AMA collects SecurityEvent logs (failed logons, successful logons, account changes).
- Queries in KQL are used to detect events of interest.
- Workflow includes health check, triage, enrichment, containment, and documentation.

## **4. Step-by-Step Explanation**

### **1. Lab Setup**

- Created a resource group and named it Tier1\_SOC
- Created a Log Analytics workspace that collects, stores, and analyzes logs and telemetry from connected resources for monitoring, querying, and security analysis, which is connected to the VM I created(Virtual Machine).
- Installed/configured AMA, which runs on your VM or server and securely collects telemetry (logs, metrics, events) from the machine and sends it to your Log Analytics workspace for analysis with the help of the DCR(Data Collection Rule), which tells the AMA what data to collect and where to send it.
- Enabled audit policy for failed logons in PowerShell.

Microsoft Cloud menu

Tier1.SOC Resource group

Help me generate Terraform for this resource group configuration. How to manage these changes more efficiently with deployment tool? Generate Bicep code to duplicate this resource group.

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete

Group by none

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

| Name 1                                            | Type                    | Location           |
|---------------------------------------------------|-------------------------|--------------------|
| DCR-AMA                                           | Data collection rule    | South Africa North |
| SecurityInsights(soc-lab-workspace)               | Solution                | South Africa North |
| SOC-Lab-Workspace                                 | Log Analytics workspace | South Africa North |
| Tier1-VM                                          | Virtual machine         | South Africa North |
| Tier1-VM-ip                                       | Public IP address       | South Africa North |
| Tier1-VM-nsg                                      | Network security group  | South Africa North |
| Tier1-VM-vnet                                     | Virtual network         | South Africa North |
| Tier1-vm780_z1                                    | Network interface       | South Africa North |
| Tier1-VM_OsDisk_1_Bed8Bc13ab0490482f1c14611268376 | Disk                    | South Africa North |

Showing 1 - 9 of 9. Display count: auto

Give feedback

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Microsoft Sentinel

Microsoft Sentinel | Content hub

Selected workspace: 'soc-lab-workspace'

Search

Refresh Install/Update Delete SIEM Migration

General

Threat management

Content management

Content hub

Repositories

Community

Configuration

425 Solutions

322 Standalone contents

0 Installed

0 Updates

Didn't find what you were looking for. Try refining your search.

security event

Status: All Content type: All Category: All Content: All

Content title

Windows Security Events

Microsoft Provider 3.0.9 Version

Support

Windows Security Events via AMA Connector. Legacy connector uses the Log Analytics agent which is about to be deprecated by **Aug 31, 2024**, and thus should only be installed where AMA is not supported.

Data Connectors: 2, Workbooks: 2, Analytic Rules: 20, Hunting Queries: 50

Learn more about Microsoft Sentinel | Learn more about Solutions

Content type

20 Data connector

2 Analytics rule

2 Workbook

50 Hunting query

Install View details

Showing 1 to 20 of 25 results.

Home > Windows Security Events >

Windows Security Events via AMA

To integrate with Windows Security Events via AMA make sure you have:

- Workspace data sources: read and write permissions.
- To collect data from non-Azure VMs, they must have Azure Arc installed and enabled. [Learn more](#)

Configuration

Enable data collection rule

Security Events logs are collected only from **Windows** agents.

Refresh

| Rule name  | Created by | Filter name |
|------------|------------|-------------|
| No results |            |             |

+Create data collection rule

## Create Data Collection Rule

Data collection rule management

[Basic](#) [Resources](#) [Collect](#) [Review + create](#)

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.

**Rule details**

Rule name \*

DRC-RULE

Subscription \* ⓘ

Azure subscription 1

Resource group \* ⓘ

Tier1\_SOC

Next: Resources >

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22631.5909]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>auditpol /set /subcategory:"Logon" /failure:enable
The command was successfully executed.

C:\Windows\System32>
```

## 2. Event Generation

- Generated failed logons manually (wrong password attempts) and observed successful logons.

Windows Security

Your credentials did not work

The credentials that were used to connect to 172.209.208.188 did not work. Please enter new credentials.

tumza4635@gmail.com

Password

MicrosoftAccount\tumza4635@gmail.com

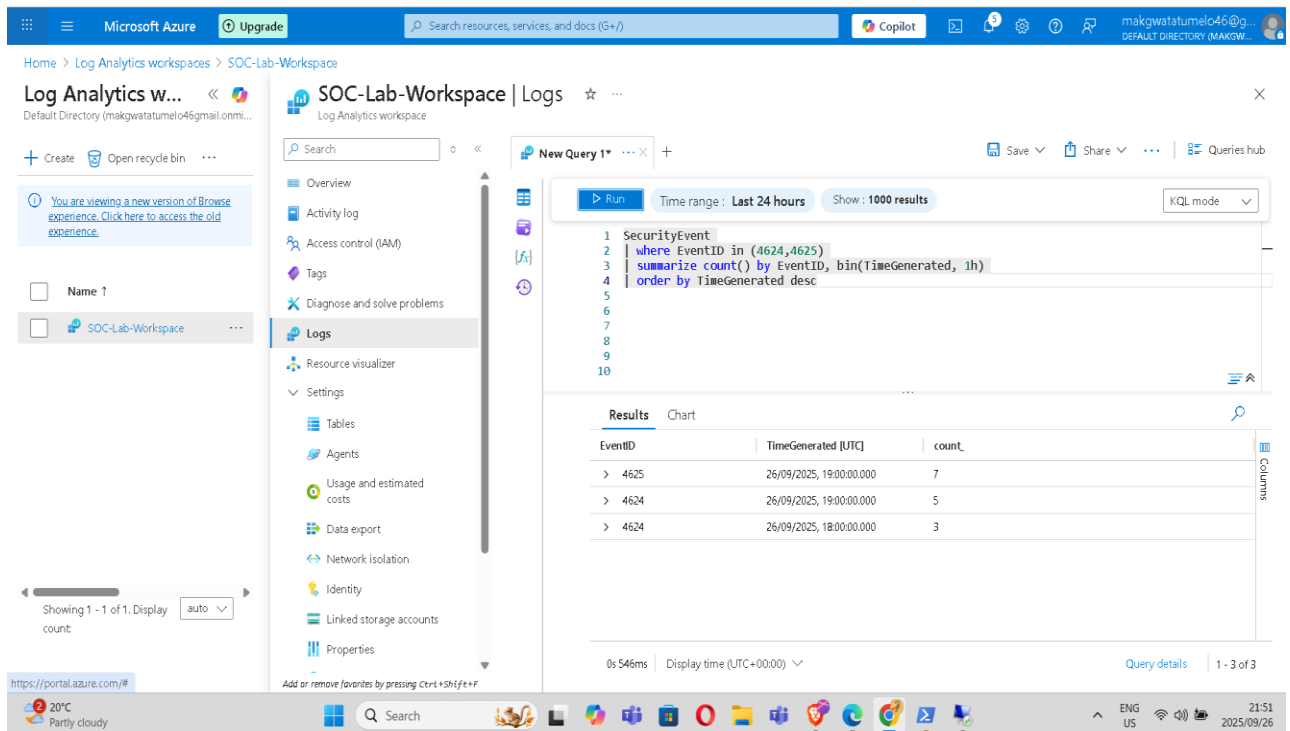
☐ Remember me

The logon attempt failed

[More choices](#)

OK Cancel

### 3. Querying Logs



Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Log Analytics workspaces > SOC-Lab-Workspace

## Log Analytics workspace

Default Directory (makgwatatumelo46@gmail.com...)

Create | Open recycle bin | ...

You are viewing a new version of Browse experience. Click here to access the old experience.

Name 1

SOC-Lab-Workspace

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Logs

Resource visualizer

Settings

Tables

Agents

Usage and estimated costs

Data export

Network isolation

Identity

Linked storage accounts

Properties

Showing 1 - 1 of 1. Display count: auto

https://portal.azure.com/#

20°C Partly cloudy

Search

SOC-Lab-Workspace | Logs

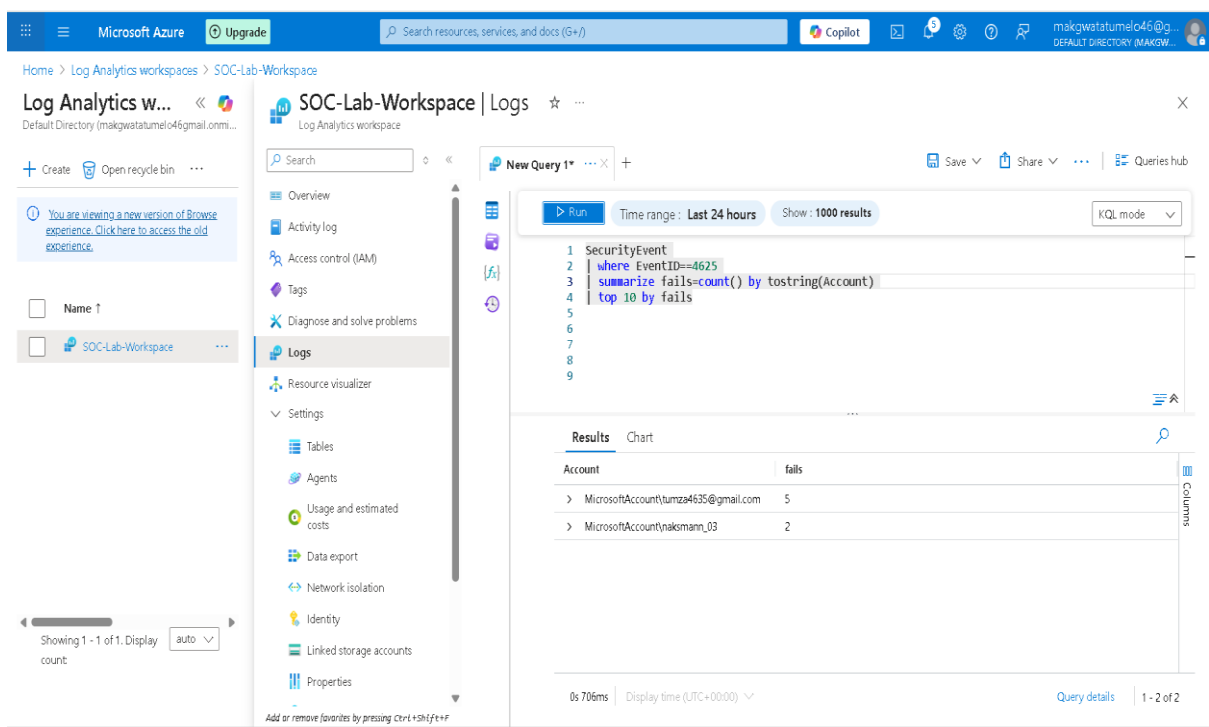
New Query 1\*

Run | Time range: Last 24 hours | Show: 1000 results | KQL mode

```
1 SecurityEvent
2 where EventID in (4624,4625)
3 summarize count() by EventID, bin(TimeGenerated, 1h)
4 order by TimeGenerated desc
```

| EventID | TimeGenerated [UTC]      | count_ |
|---------|--------------------------|--------|
| > 4625  | 26/09/2025, 19:00:00.000 | 7      |
| > 4624  | 26/09/2025, 19:00:00.000 | 5      |
| > 4624  | 26/09/2025, 18:00:00.000 | 3      |

0s 546ms | Display time (UTC+00:00) | Query details | 1 - 3 of 3



Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Log Analytics workspaces > SOC-Lab-Workspace

## Log Analytics workspace

Default Directory (makgwatatumelo46@gmail.com...)

Create | Open recycle bin | ...

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Network isolation

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Linked storage accounts

Properties

Showing 1 - 1 of 1. Display count: auto

https://portal.azure.com/#

20°C Partly cloudy

Search

SOC-Lab-Workspace | Logs

New Query 1\*

Run | Time range: Last 24 hours | Show: 1000 results | KQL mode

```
1 SecurityEvent
2 where EventID==4625
3 summarize fails=count() by tostring(Account)
4 top 10 by fails
```

| Account                                | fails |
|----------------------------------------|-------|
| > MicrosoftAccount(tumza4635@gmail.com | 5     |
| > MicrosoftAccount(naksmann_03         | 2     |

0s 706ms | Display time (UTC+00:00) | Query details | 1 - 2 of 2

## 4. Monitoring & Triage

- Checked VM heartbeat for health monitoring.
- Identified failed logon patterns and repeated attempts.

The screenshot displays the Microsoft Azure Log Analytics workspace interface. The top navigation bar includes the Microsoft Azure logo, an 'Upgrade' button, a search bar, and a Copilot icon. The main header shows 'Log Analytics w...' and 'SOC-Lab-Workspace | Logs'. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Logs, Resource visualizer, Settings, Tables, Agents, Usage and estimated costs, Data export, Network isolation, Identity, Linked storage accounts, and Properties. The 'Logs' section is currently selected. The main content area shows a KQL query editor with the following query:

```
1 Heartbeat
2 where TimeGenerated > ago(30m)
3 project TimeGenerated, Computer, OSType
4 sort by TimeGenerated desc
5
6
```

The query results are displayed in a table with the following columns: TimeGenerated [UTC], Computer, and OSType. The results show a list of heartbeat events for 'Tier1-VM' on 'Windows' operating systems, sorted by TimeGenerated in descending order.

| TimeGenerated [UTC]        | Computer | OSType  |
|----------------------------|----------|---------|
| > 26/09/2025, 20:10:11.973 | Tier1-VM | Windows |
| > 26/09/2025, 20:09:11.958 | Tier1-VM | Windows |
| > 26/09/2025, 20:08:11.958 | Tier1-VM | Windows |
| > 26/09/2025, 20:07:11.942 | Tier1-VM | Windows |
| > 26/09/2025, 20:06:11.942 | Tier1-VM | Windows |
| > 26/09/2025, 20:05:11.927 | Tier1-VM | Windows |

The bottom of the interface shows the query execution time (3s 117ms) and the display time (UTC+00:00). The URL in the bottom left corner is 'https://portal.azure.com/#'.

## 5. Enrichment

- Documented account names, computers, logon types, and timestamps.
- Added context for incident reporting.

## 5. Findings / Analysis

- **What happened?**
  - Multiple failed logon attempts were detected in the lab.
  - Successful logons were recorded for comparison.
- **Evidence**
  - Screenshots of KQL query results showing failed vs successful logons.
  - Logs with EventID 4625 (failed) and 4624 (successful).
  - Entity details: Account names, computers, and logon types.

The screenshot displays the Microsoft Azure portal interface. At the top, there's a navigation bar with 'Microsoft Azure' and 'Upgrade' buttons. Below it, the breadcrumb trail shows 'Home > Log Analytics workspaces > SOC-Lab-Workspace'. The main content area is titled 'SOC-Lab-Workspace | Logs'. On the left, a sidebar lists various Log Analytics features like Overview, Activity log, Access control (AM), Tags, Diagnose and solve problems, Logs, Resource visualizer, Settings, Tables, Agents, Usage and estimated costs, Data export, Network isolation, Identity, Linked storage accounts, and Properties. The 'Logs' section is currently selected. In the center, a KQL query is shown in a text editor:

```
1 SecurityEvent
2 where TimeGenerated > ago(30m)
3 where EventID == 4625
4 project TimeGenerated, Computer, Account, LogonType, EventID
5 sort by TimeGenerated desc
```

Below the query editor, the 'Results' tab is active, displaying a table with the following data:

| TimeGenerated [UTC]        | Computer | Account                      | LogonType | EventID |
|----------------------------|----------|------------------------------|-----------|---------|
| > 26/09/2025, 20:10:11.973 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |
| > 26/09/2025, 20:09:11.958 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |
| > 26/09/2025, 20:08:11.958 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |
| > 26/09/2025, 20:07:11.942 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |
| > 26/09/2025, 20:06:11.942 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |
| > 26/09/2025, 20:05:11.927 | Tier1-VM | MicrosoftAccount\nakemann_03 | 3         | 4625    |

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 21:49 on 2025/09/26.

## • Actions Taken

- Containment (lab simulation): documented potential actions (lock account, reset password).
- Enrichment: linked related events in Sentinel and added comments.

## • Root Cause

- Failed logons were intentionally generated for lab purposes.

## • Lessons Learned & Prevention

- Importance of monitoring SecurityEvent logs.
- Practice triage workflow: health check → alert validation → enrichment → documentation.
- Analytics rules can help detect repeated failed logons in production.



## 6. Use-Case Sheet (Tier-1 Version)

| Field               | Entry                                                                       |
|---------------------|-----------------------------------------------------------------------------|
| Trigger             | SecurityEvent with EventID 4625                                             |
| Inputs              | Account, Computer, TimeGenerated                                            |
| Steps               | 1. Query log → 2. Add comment to incident → 3. Document in the lab workbook |
| External Calls      | Optional: Threat intelligence enrichment (OTX/VT)                           |
| Rate Limits         | N/A (lab)                                                                   |
| Secrets Storage     | N/A (lab)                                                                   |
| Outputs to Incident | Incident created with failed logon count                                    |
| Error Handling      | Log query failure; skip if no events                                        |

## 7. Playbook Design Doc (Tier-1 Version)

| Field               | Entry                                                                       |
|---------------------|-----------------------------------------------------------------------------|
| Trigger             | SecurityEvent with EventID 4625                                             |
| Inputs              | Account, Computer, TimeGenerated                                            |
| Steps               | 1. Query log → 2. Add comment to incident → 3. Document in the lab workbook |
| External Calls      | Optional: Threat intelligence enrichment (OTX/VT)                           |
| Rate Limits         | N/A (lab)                                                                   |
| Secrets Storage     | N/A (lab)                                                                   |
| Outputs to Incident | Incident created with failed logon count                                    |
| Error Handling      | Log query failure; skip if no events                                        |

## 8. Shift Handover (Tier-1 Version)

| Field                 | Entry                                                         |
|-----------------------|---------------------------------------------------------------|
| Open Incidents        | Failed logons detected in the last 24h                        |
| Pending Actions       | Documented lab containment steps                              |
| Risk Watchlist        | Repeated failed logons                                        |
| Connector Health      | AMA agent running; VM heartbeat verified                      |
| Notable Events        | Any unusual repeated logons                                   |
| Next Shift Priorities | Continue monitoring logs; practice triage on new test events. |

## 9. Conclusion

- Tier-1 SOC tasks can be practiced in a lab using SecurityEvent logs and Sentinel.
- The lab demonstrated monitoring, triage, enrichment, and documentation workflows.
- SecurityEvent logs are protected; only real system events (failed logons) can be generated.
- The lab reinforces structured workflows, documentation, and readiness for incident escalation.